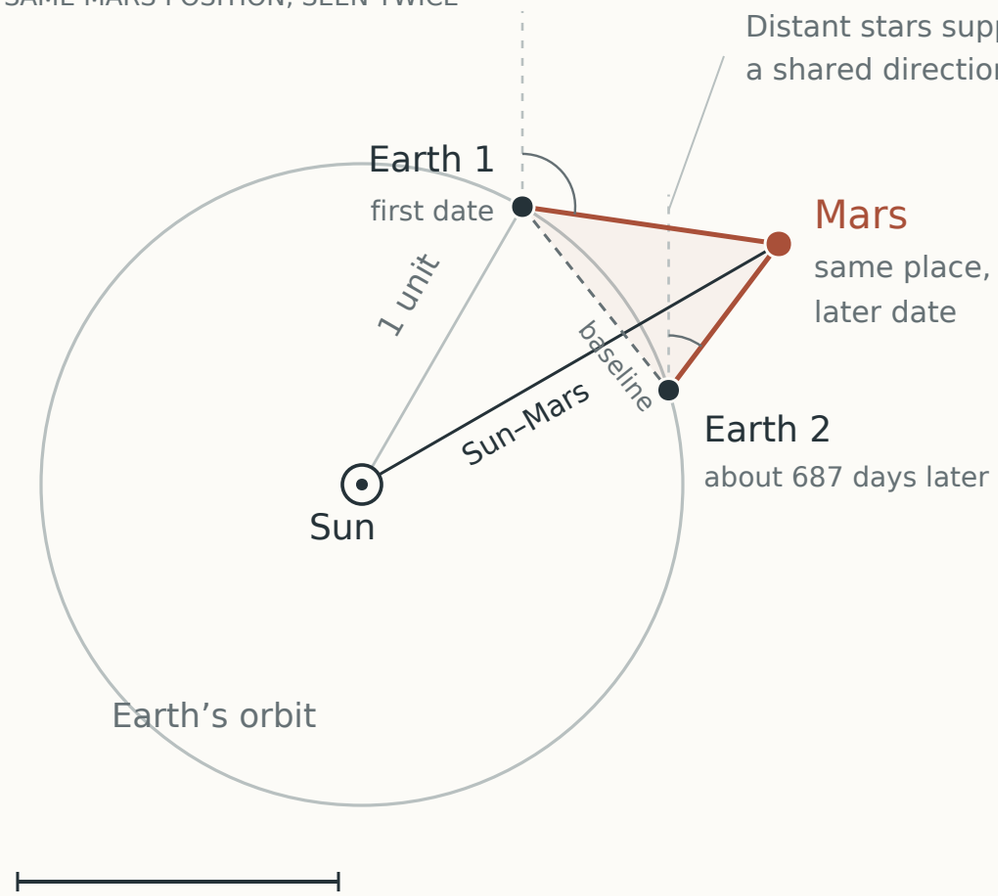
How Kepler found Mars without going there

One returning planet. Two observing places. A triangle with a scale.

THE SAME MARS POSITION, SEEN TWICE



1 unit · mean Earth–Sun distance

Teaching circle only. Kepler also reconstructed and refined Earth’s orbit.

1 Wait for Mars to return.

Tycho’s records supplied paired sightings about 687 days apart: one Mars orbit, not the interval between successive oppositions. Earth is now elsewhere.

2 Cross the two sightlines.

Place both Earth locations on a map of Earth’s orbit. Draw each measured direction to Mars. Their intersection recovers this one Mars position.

3 Read a relative distance.

The Earth–Earth chord sets the scale. Measure Sun to Mars on that map: about 1.5 Earth–Sun units here. That distance varies around the orbit.

A star direction alone is not a distance.

Use a true triangle, not three points in a line.

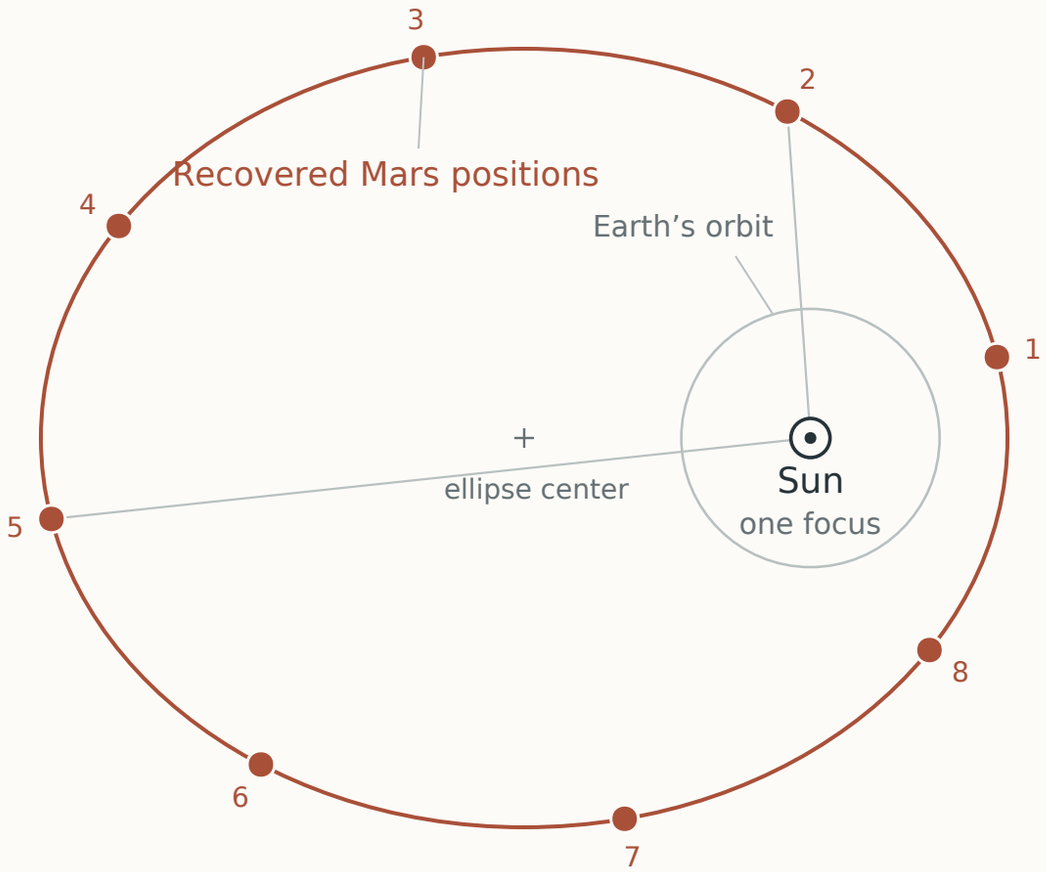
Sources: “Using Kepler’s Triangulation Technique,” Univ. of Central Arkansas teaching handout, pp. 2–3 (faculty.uca.edu/saustin/3110/mars.pdf).

History: O. Gingerich, “The great Martian catastrophe and how Kepler fixed it,” Physics Today 64(9), 50–54 (2011). • Prepared 7 Nov 2026.

Many triangles reveal an ellipse

Faria’s lesson: locate the points first; let the evidence choose the curve.

REPEAT AT OTHER PLACES IN MARS’S ORBIT

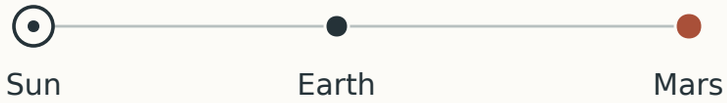


Each dot represents a different repeat-position observation pair.

Ellipse flattening and focus offset exaggerated; this panel is not to scale.

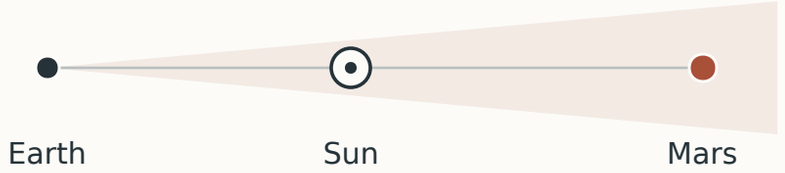
TWO ALIGNMENTS — NOT TWO DISTANCE METHODS

Opposition: Earth is in the middle.



Mars is opposite the Sun in Earth’s sky.
The straight alignment alone does not fix distance.

Conjunction: the Sun is in the middle.



Mars lies in the Sun’s direction: solar glare prevents a usable ordinary sighting near conjunction.

No permanently hidden half-orbit.

Visibility changes with Earth’s viewpoint.
Mars can be seen in many orbital places over time.

Sources: NASA Science, “Orbits and Kepler’s Laws”; “Close Encounter: Mars at Opposition”; “Geometry of Mars Solar Conjunction.”

science.nasa.gov • Geometry simplified to one plane. Dots, bodies and glare cone are illustrative; no historical observations are plotted.